```
import numpy as np
import pandas as pd

df = pd.read_csv("/content/AB_NYC_2019.csv")
df.head()
```

	id	name	host_id	host_name	neighbourhood_group	neighbourhood	latitude	longitude	room_type	price	minimum_nights
0	2539	Clean & quiet apt home by the park	2787	John	Brooklyn	Kensington	40.64749	-73.97237	Private room	149	1
1	2595	Skylit Midtown Castle	2845	Jennifer	Manhattan	Midtown	40.75362	-73.98377	Entire home/apt	225	1
2	3647	THE VILLAGE OF HARLEM....NEW YORK !	4632	Elisabeth	Manhattan	Harlem	40.80902	-73.94190	Private room	150	3
3	3831	Cozy Entire Floor of Brownstone	4869	LisaRoxanne	Brooklyn	Clinton Hill	40.68514	-73.95976	Entire home/apt	89	1
4	5022	Entire Apt: Spacious Studio/Loft by central park	7192	Laura	Manhattan	East Harlem	40.79851	-73.94399	Entire home/apt	80	10

```
df.isnull().sum()
```

```
id                                    0
name                                 16
host_id                               0
host_name                            21
neighbourhood_group                   0
neighbourhood                         0
latitude                              0
longitude                             0
room_type                             0
price                                 0
minimum_nights                        0
number_of_reviews                     0
last_review                       10052
reviews_per_month                 10052
calculated_host_listings_count        0
availability_365                      0
dtype: int64
```

```
df.fillna({'reviews_per_month':0},inplace=True)
```

```
df.isnull().sum()
```

```
id                                    0
name                                 16
host_id                               0
host_name                            21
neighbourhood_group                   0
neighbourhood                         0
latitude                              0
longitude                             0
room_type                             0
price                                 0
minimum_nights                        0
number_of_reviews                     0
last_review                       10052
reviews_per_month                     0
calculated_host_listings_count        0
availability_365                      0
dtype: int64
```

```
df.drop(['name', 'last_review'], axis=1, inplace=True)
```

```
df.isnull().sum()
```

```
id                                 0
host_id                            0
host_name                         21
neighbourhood_group                0
neighbourhood                      0
latitude                           0
longitude                          0
room_type                          0
price                              0
minimum_nights                     0
number_of_reviews                  0
reviews_per_month                  0
calculated_host_listings_count     0
availability_365                   0
dtype: int64
```

```
df.head(20)
```

	id	host_id	host_name	neighbourhood_group	neighbourhood	latitude	longitude	room_type	price	minimum_nights	number_of_revi
0	2539	2787	John	Brooklyn	Kensington	40.64749	-73.97237	Private room	149	1	
1	2595	2845	Jennifer	Manhattan	Midtown	40.75362	-73.98377	Entire home/apt	225	1	
2	3647	4632	Elisabeth	Manhattan	Harlem	40.80902	-73.94190	Private room	150	3	
3	3831	4869	LisaRoxanne	Brooklyn	Clinton Hill	40.68514	-73.95976	Entire home/apt	89	1	
4	5022	7192	Laura	Manhattan	East Harlem	40.79851	-73.94399	Entire home/apt	80	10	
5	5099	7322	Chris	Manhattan	Murray Hill	40.74767	-73.97500	Entire home/apt	200	3	
6	5121	7356	Garon	Brooklyn	Bedford-Stuyvesant	40.68688	-73.95596	Private room	60	45	
7	5178	8967	Shunichi	Manhattan	Hell's Kitchen	40.76489	-73.98493	Private room	79	2	
8	5203	7490	MaryEllen	Manhattan	Upper West Side	40.80178	-73.96723	Private room	79	2	
9	5238	7549	Ben	Manhattan	Chinatown	40.71344	-73.99037	Entire home/apt	150	1	
10	5295	7702	Lena	Manhattan	Upper West Side	40.80316	-73.96545	Entire home/apt	135	5	
11	5441	7989	Kate	Manhattan	Hell's Kitchen	40.76076	-73.98867	Private room	85	2	
12	5803	9744	Laurie	Brooklyn	South Slope	40.66829	-73.98779	Private room	89	4	
13	6021	11528	Claudio	Manhattan	Upper West Side	40.79826	-73.96113	Private room	85	2	
14	6090	11975	Alina	Manhattan	West Village	40.73530	-74.00525	Entire home/apt	120	90	
15	6848	15991	Allen & Irina	Brooklyn	Williamsburg	40.70837	-73.95352	Entire home/apt	140	2	
16	7097	17571	Jane	Brooklyn	Fort Greene	40.69169	-73.97185	Entire home/apt	215	2	
17	7322	18946	Doti	Manhattan	Chelsea	40.74192	-73.99501	Private room	140	1	
18	7726	20950	Adam And Charity	Brooklyn	Crown Heights	40.67592	-73.94694	Entire home/apt	99	3	
19	7750	17985	Sing	Manhattan	East Harlem	40.79685	-73.94872	Entire home/apt	190	7	

```
df.to_csv('AB_NYC_2019_updated.csv',index= False)
```

```
data_original = pd.read_csv('/content/AB_NYC_2019_updated.csv')
data_original.head()
```

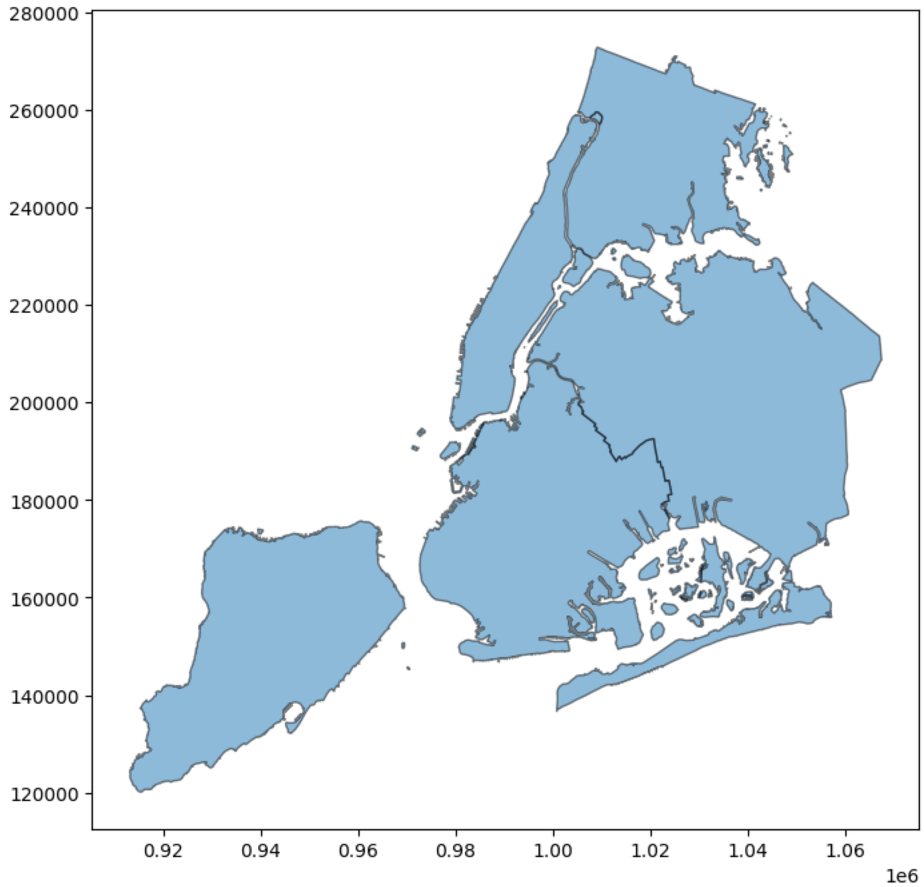
	id	host_id	host_name	neighbourhood_group	neighbourhood	latitude	longitude	room_type	price	minimum_nights	number_of_revie
0	2539	2787	John	Brooklyn	Kensington	40.64749	-73.97237	Private room	149	1	
1	2595	2845	Jennifer	Manhattan	Midtown	40.75362	-73.98377	Entire home/apt	225	1	
2	3647	4632	Elisabeth	Manhattan	Harlem	40.80902	-73.94190	Private room	150	3	
3	3831	4869	LisaRoxanne	Brooklyn	Clinton Hill	40.68514	-73.95976	Entire home/apt	89	1	2
4	5022	7192	Laura	Manhattan	East Harlem	40.79851	-73.94399	Entire home/apt	80	10	

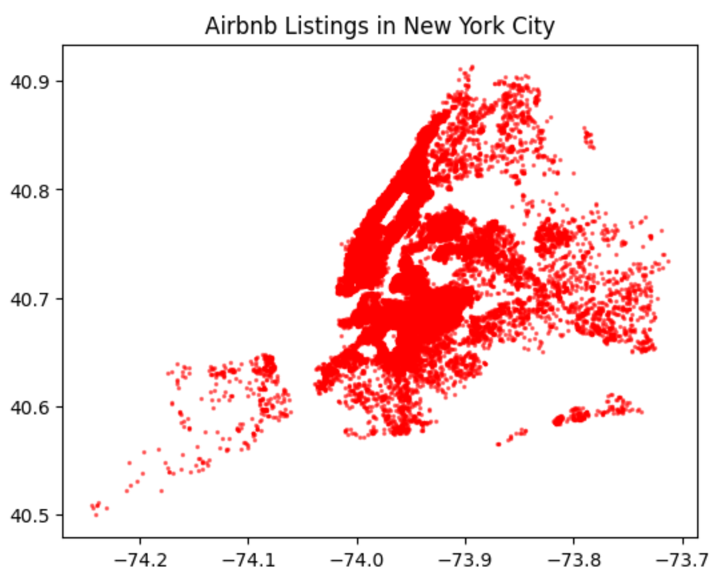
```
print(df[['latitude', 'longitude']].describe())
```

```
           latitude     longitude
count  48895.000000  48895.000000
mean      40.728949    -73.952170
std        0.054530      0.046157
min       40.499790    -74.244420
25%       40.690100    -73.983070
50%       40.723070    -73.955680
75%       40.763115    -73.936275
max       40.913060    -73.712990
```

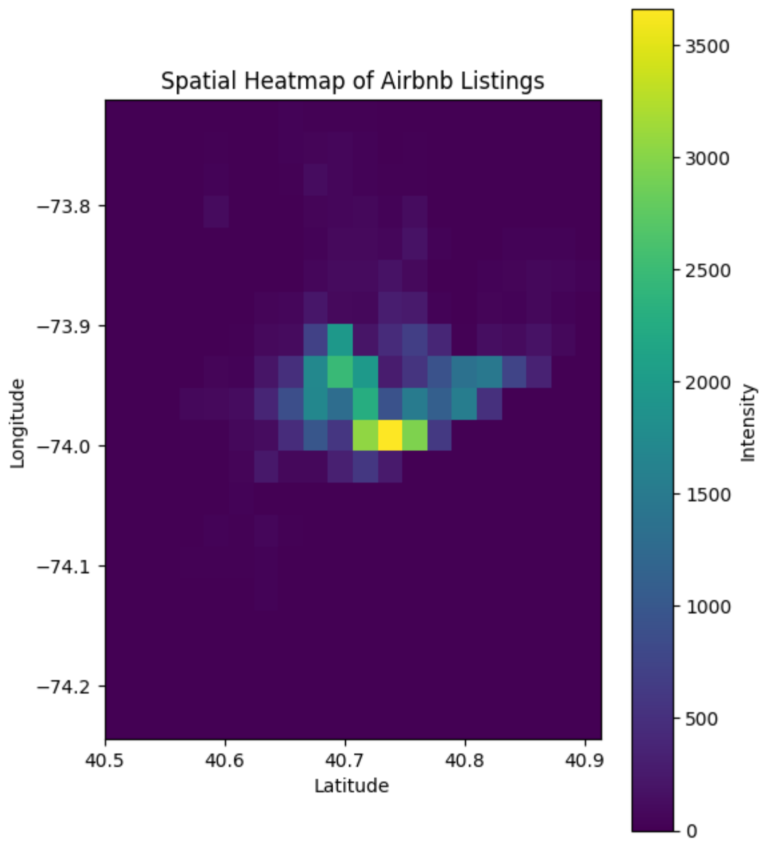
```
import geopandas as gpd
import matplotlib.pyplot as plt

gdf = gpd.GeoDataFrame(data_original, geometry=gpd.points_from_xy(data_original.longitude, data_original.latitude))

# Load a basemap of New York City using the geopandas.datasets package
nyc = gpd.read_file(gpd.datasets.get_path('nybb'))
ax = nyc.plot(figsize=(8, 8), alpha=0.5, edgecolor='k')

# Plot Airbnb listings on top of the basemap
gdf.plot(color='red', alpha=0.5, markersize=2)

plt.title('Airbnb Listings in New York City')
plt.show()
```

```
<ipython-input-23-91f4d0366a0f>:7: FutureWarning: The geopandas.dataset module is deprecated and will be removed in GeoPandas 1.0. You c

from geodatasets import get_path
path_to_file = get_path('nybb')

  nyc = gpd.read_file(gpd.datasets.get_path('nybb'))
```



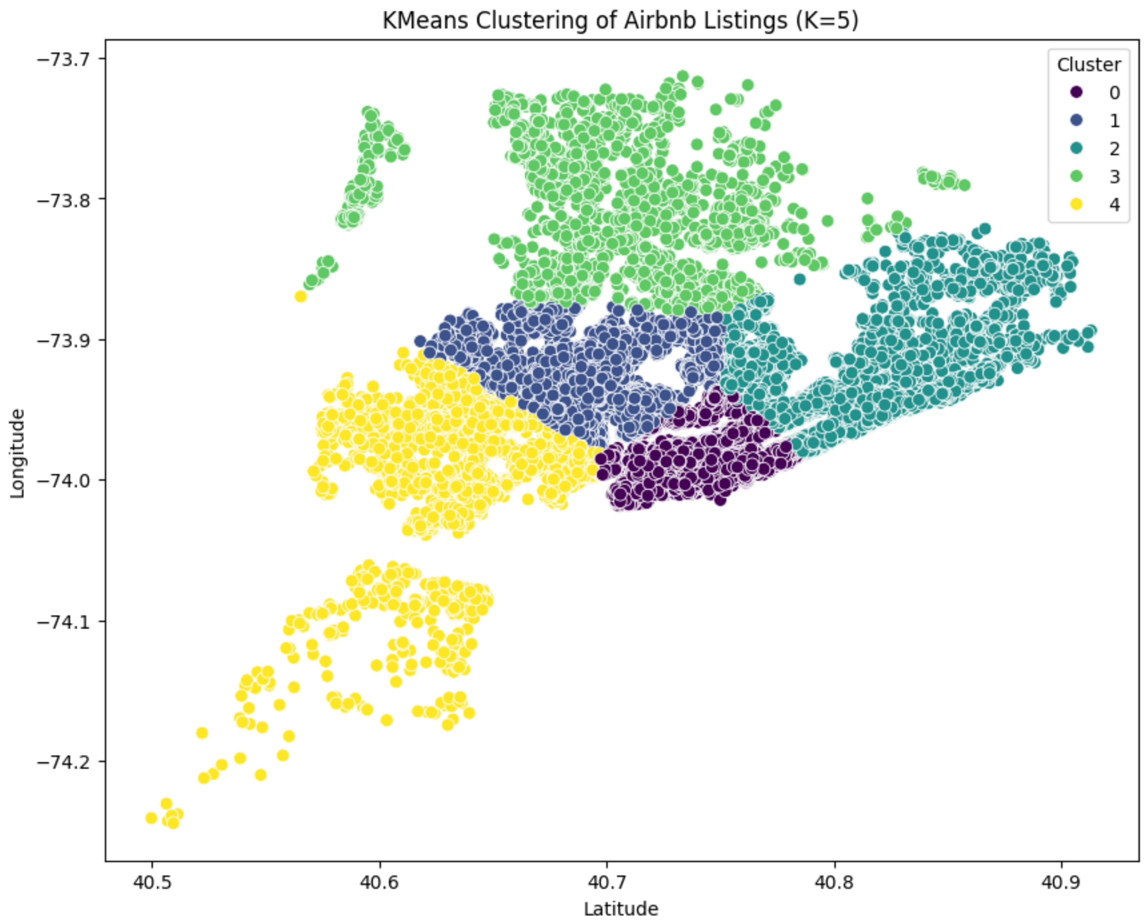


```
import seaborn as sns


# Create a 2D histogram (heatmap)
heatmap, xedges, yedges = np.histogram2d(data_original['latitude'], data_original['longitude'], bins=(20, 20))
extent = [xedges[0], xedges[-1], yedges[0], yedges[-1]]

# Plot the heatmap
plt.figure(figsize=(6, 8))
plt.imshow(heatmap.T, extent=extent, origin='lower', cmap='viridis')
plt.colorbar(label='Intensity')
plt.title('Spatial Heatmap of Airbnb Listings')
plt.xlabel('Latitude')
plt.ylabel('Longitude')
plt.show()
```



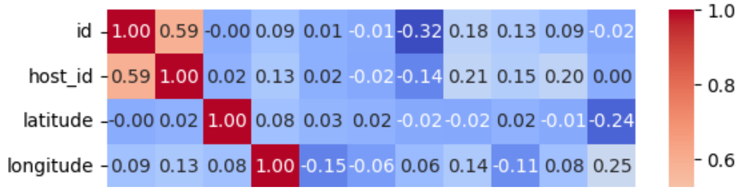

```
from sklearn.cluster import KMeans


num_clusters = 5

# Apply KMeans clustering
kmeans = KMeans(n_clusters=num_clusters, random_state=50)
data_original['cluster'] = kmeans.fit_predict(data_original[['latitude', 'longitude']])

# Visualize clusters on a scatter plot
plt.figure(figsize=(10, 8))
sns.scatterplot(x='latitude', y='longitude', hue='cluster', data=data_original, palette='viridis', s=50)
plt.title(f'KMeans Clustering of Airbnb Listings (K={num_clusters})')
plt.xlabel('Latitude')
plt.ylabel('Longitude')
plt.legend(title='Cluster')
plt.show()
```

```
/usr/local/lib/python3.10/dist-packages/sklearn/cluster/_kmeans.py:870: FutureWarning: The default value of `n_init` will change from 10
  warnings.warn(
```




```
#Correlation analysis

correlation_matrix = data_original.corr()

# Heatmap visualization
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt=".2f")
plt.show()
```

```
<ipython-input-19-1d3c4eb6b26a>:3: FutureWarning: The default value of numeric_only in DataFrame.corr is deprecated. In a future version
  correlation_matrix = data_original.corr()
```




```
# Create a heatmap
plt.figure(figsize=(8, 8))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt=".3f")
plt.title('Correlation Matrix')
plt.show()
```